

DATA VISUALIZATION DOCUMENTATION

Matplotlib and Seaborn

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Beginner Level Task – Visualization Library Documentation

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Data Visualization Documentation

(Matplotlib and Seaborn)

1. Introduction:

Data visualization is the graphical representation of data using charts and graphs. It helps users understand patterns, trends, and relationships within data.

Python provides several visualization libraries such as Matplotlib, Seaborn, Plotly, and Bokeh. In this documentation, we explore Matplotlib and Seaborn, two widely used libraries in Data Science.

2. Matplotlib Overview:

Matplotlib is one of the most popular Python libraries for data visualization. It provides complete control over graph design and supports various chart types.

Key Features:

- Easy to learn
- Highly customizable
- Supports static, animated, and interactive graphs
- Integrates well with NumPy and Pandas

3. Matplotlib Graph Types:

- **Line Plot:** Used to show trends over time.

Example: Monthly Sales Analysis

Code:

```
import matplotlib.pyplot as plt
```

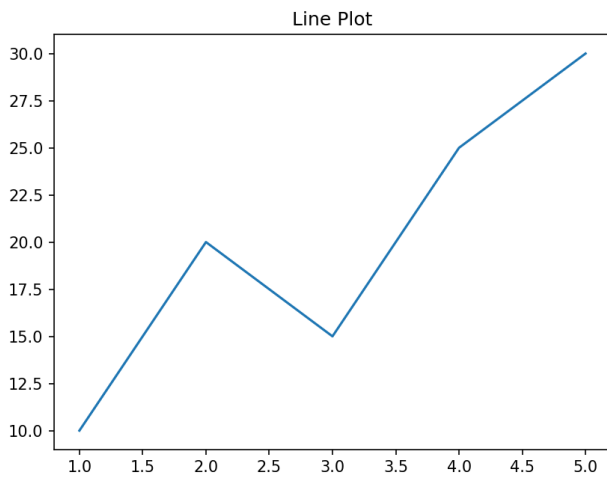
```
x = [1,2,3,4,5]
```

```
y = [10,20,15,25,30]
```

```
plt.plot(x,y)
```

```
plt.title("Line Plot")
```

```
plt.show()
```



- **Bar Chart:** Compares different categories.

Example: Comparing student marks.

Code:

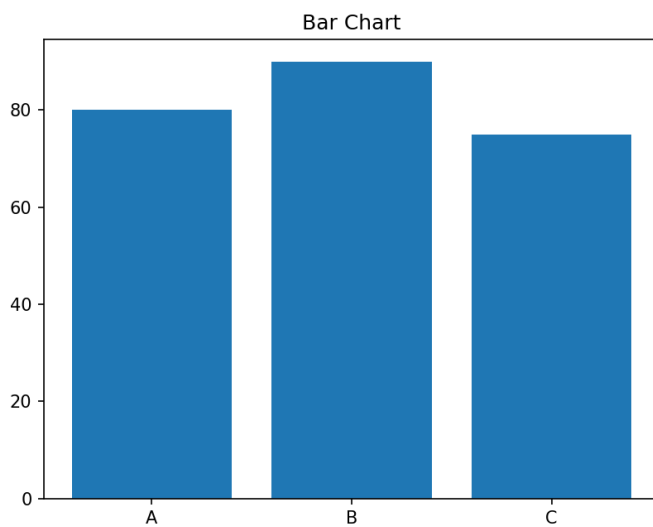
```
students = ["A","B","C"]
```

```
marks = [80,90,75]
```

```
plt.bar(students, marks)
```

```
plt.title("Bar Chart")
```

```
plt.show()
```



- **Scatter Plot:** Shows relationship between two variables.

Use Case: Study hours vs marks.

Code:

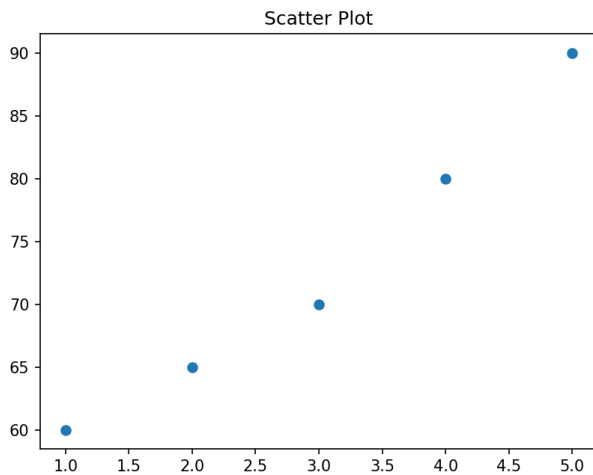
```
x = [1,2,3,4,5]
```

```
y = [60,65,70,80,90]
```

```
plt.scatter(x,y)
```

```
plt.title("Scatter Plot")
```

```
plt.show()
```



- Histogram: Shows frequency distribution.

Use Case: Age distribution of students.

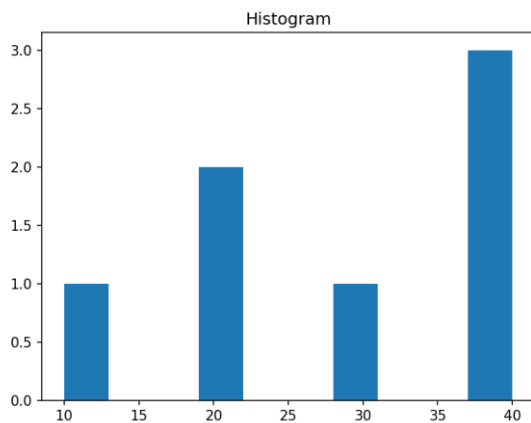
Code:

```
data = [10,20,20,30,40,40,40]
```

```
plt.hist(data)
```

```
plt.title("Histogram")
```

```
plt.show()
```



- Pie Chart: Shows percentage contribution.

Use Case: Market share analysis.

Code:

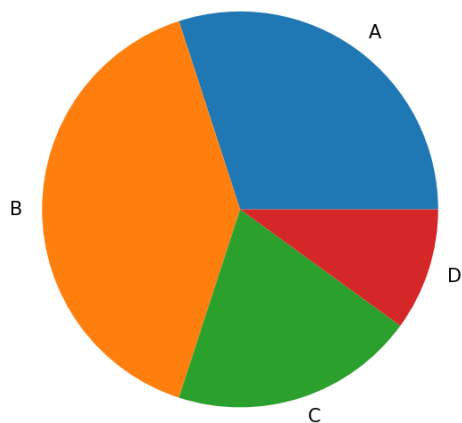
```
import matplotlib.pyplot as plt
```

```
sizes = [30,40,20,10]
```

```
labels = ["A","B","C","D"]
```

```
plt.pie(sizes, labels=labels)
```

```
plt.show()
```



4. Seaborn Overview:

Seaborn is a Python data visualization library built on top of Matplotlib.

It provides beautiful default themes and statistical visualizations with less code.

Key Features:

- Attractive default styles
- Statistical plotting support
- Easy integration with Pandas
- Simple syntax

5. Seaborn Graph Types:

- Line Plot:

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

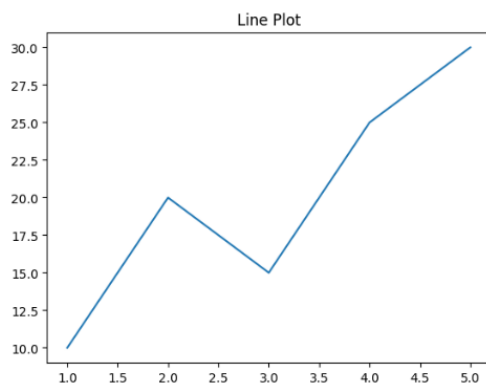
```
x=[1,2,3,4,5]
```

```
y=[10,20,15,25,30]
```

```
sns.lineplot(x=x,y=y)
```

```
plt.title("Line Plot")
```

```
plt.show()
```



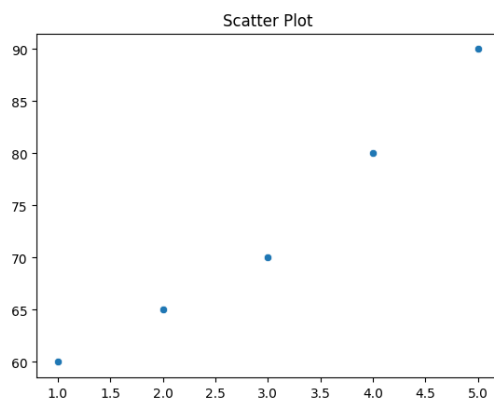
- Scatter Plot:

```
sns.scatterplot(x=[1,2,3,4,5],
```

```
y=[60,65,70,80,90])
```

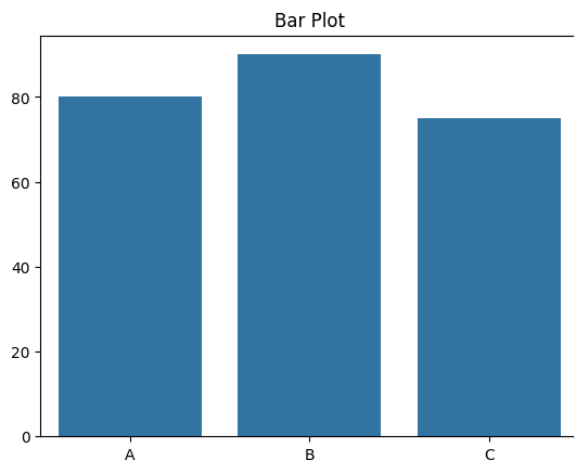
```
plt.title("Scatter Plot")
```

```
plt.show()
```



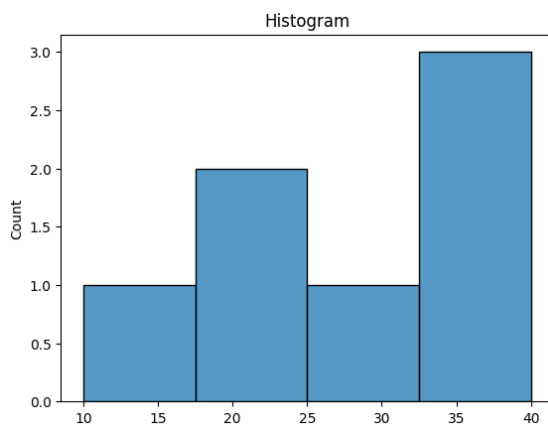
- Bar Plot:

```
sns.barplot(x=["A","B","C"],  
            y=[80,90,75])  
  
plt.title("Bar Plot")  
  
plt.show()
```



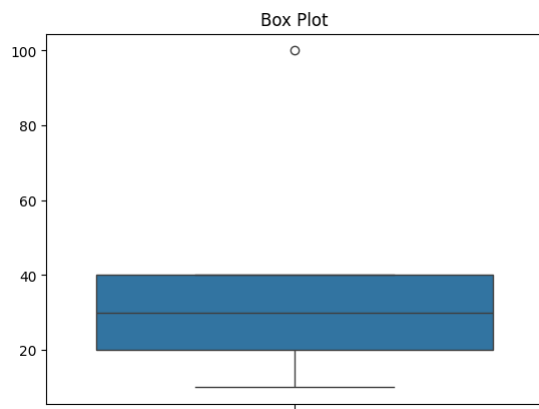
- Histogram:

```
sns.histplot([10,20,20,30,40,40,40])  
  
plt.title("Histogram")  
  
plt.show()
```



- Box Plot:

```
sns.boxplot(data=[10,20,30,40,100])  
  
plt.title("Box Plot")  
  
plt.show()
```

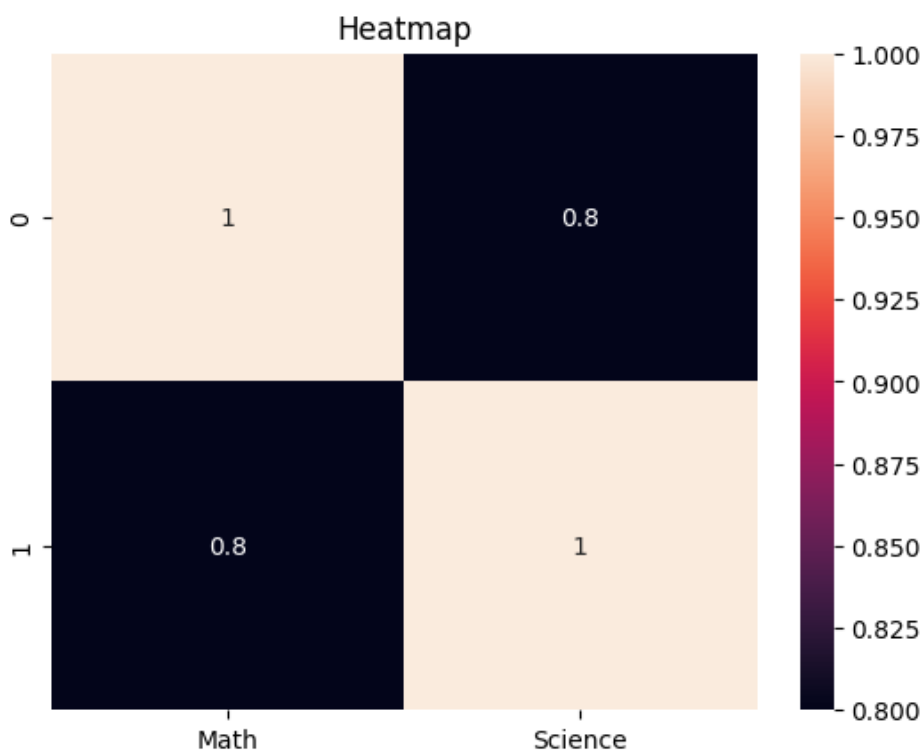


- Heatmap:

```
import pandas as pd
```

```
data = pd.DataFrame(
    [[1,0.8],[0.8,1]],
    columns=["Math","Science"]
)
```

```
sns.heatmap(data, annot=True)
plt.title("Heatmap")
plt.show()
```



6. Comparison:

Feature	Matplotlib	Seaborn
Ease of Use	Moderate	Easy
Customization	Very High	High
Appearance	Basic	Attractive
Statistical Graphs	Limited	Excellent
Learning Curve	Higher	Lower
Large Dataset Handling	Good	Good

7. Conclusion:

- Matplotlib and Seaborn are powerful visualization libraries in Python.
- Matplotlib provides extensive customization and control, while Seaborn
- offers attractive statistical visualizations with simpler syntax.
- Together, they are essential tools for data analysis and data science projects.